

ACHINSK, Russian Federation

WMO: 294670

Lat: 56.283N

Lon: 90.517E

Elev: 844

StdP: 14.25

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.6	-23.4	-34.5	0.8	-29.6	-28.3	1.2	-23.2	22.2	18.6	20.3	15.8	3.7	230	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	84.4	65.7	80.9	64.3	77.8	63.1	68.2	79.6	66.5	77.1	65.1	74.9	4.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	93.1	73.2	62.7	87.8	70.8	61.1	83.0	70.4	33.1	79.5	31.8	77.1	30.6	75.0	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.5	14.4	-31.8	89.2	9.3	3.3	-38.4	91.5	-43.8	93.4	-49.0	95.3	-55.7	97.7		
(5)			-31.1	70.6	8.7	2.1	-37.4	72.1	-42.5	73.4	-47.4	74.6	-53.8	76.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.9	4.9	11.8	23.2	37.5	51.1	63.6	66.0	61.8	48.5	34.9	16.7	9.2
	DBStd	24.09	14.83	12.39	11.44	9.29	10.62	7.95	6.05	6.77	8.52	9.85	14.58	14.39
	HDD50	6686	1398	1069	832	387	120	8	0	4	129	475	998	1266
	HDD65	10847	1863	1489	1297	824	447	116	61	142	496	933	1448	1731
	CDD50	1539	0	0	0	13	154	415	496	369	85	7	0	0
	CDD65	227	0	0	0	0	16	74	92	43	2	0	0	0
	CDH74	1671	0	0	0	2	130	605	614	306	14	0	0	0
	CDH80	383	0	0	0	0	27	151	144	60	1	0	0	0
	WSAvg	6.5	7.2	7.2	7.5	7.5	6.8	4.7	4.2	4.5	5.7	7.2	7.8	7.6
	PrecAvg	16.6	0.6	0.5	0.5	0.9	1.5	2.2	3.0	2.6	1.6	1.3	1.0	0.9
(15) Precipitation	PrecMax	24.8	1.4	1.3	1.3	2.0	4.2	5.0	6.8	5.3	3.0	3.4	2.0	2.1
	PrecMin	10.8	0.1	0.1	0.1	0.1	0.2	0.6	0.3	0.9	0.5	0.4	0.2	0.2
	PrecStd	3.5	0.3	0.3	0.2	0.5	0.9	1.2	1.7	1.2	0.7	0.7	0.4	0.4
	PrecStd	3.5	0.3	0.3	0.2	0.5	0.9	1.2	1.7	1.2	0.7	0.7	0.4	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	36.4	47.6	67.4	84.2	88.9	88.1	85.5	75.8	63.0	49.2	35.4
		MCWB	31.8	31.9	38.4	49.7	60.2	65.7	69.0	67.9	59.4	49.7	43.3	31.5
	2%	DB	30.4	32.8	43.7	61.2	77.6	84.3	83.9	81.2	70.1	56.8	42.3	32.4
		MCWB	27.3	29.1	35.9	47.5	56.9	64.9	66.9	66.0	57.6	47.6	38.0	29.5
	5%	DB	27.4	30.4	40.8	56.7	72.8	80.4	80.2	77.2	65.6	52.6	38.2	29.4
		MCWB	24.9	27.3	34.4	44.7	55.0	63.4	65.4	63.9	55.4	45.0	34.2	26.9
	10%	DB	24.0	27.9	37.6	52.0	68.6	77.1	76.9	73.7	61.4	48.6	34.6	26.1
		MCWB	21.6	25.0	32.4	42.3	53.2	62.4	64.3	62.3	53.2	42.8	31.6	23.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.3	32.6	39.8	52.0	61.6	69.1	71.0	69.9	60.9	51.3	43.7	32.2
		MCDB	34.5	36.0	45.4	65.1	79.0	81.9	82.9	82.0	71.8	58.6	48.6	34.6
	2%	WB	27.5	29.7	37.3	48.7	58.7	66.9	68.6	67.5	58.4	48.7	38.0	29.7
		MCDB	29.8	32.0	42.8	59.2	72.2	79.3	80.0	78.0	68.1	55.6	41.6	32.3
	5%	WB	25.1	27.5	34.7	45.7	56.6	65.5	67.0	65.3	56.4	46.0	34.4	27.1
		MCDB	27.5	30.2	39.9	55.2	69.3	77.3	77.2	74.9	64.2	51.8	37.8	29.4
	10%	WB	21.6	25.0	32.6	42.7	54.5	63.8	65.6	63.5	54.2	43.1	31.8	24.1
		MCDB	23.9	27.9	37.3	51.4	67.0	74.5	74.7	71.9	60.0	48.6	34.4	26.2
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	11.0	12.5	14.1	16.5	16.5	15.2	15.0	12.9	10.3	9.9	9.9
		MCDBR	11.3	10.7	14.3	20.6	21.4	22.0	20.4	20.3	20.8	15.5	10.6	9.7
		MCWBR	9.7	9.1	9.9	11.7	10.1	9.2	7.9	8.8	10.3	9.2	7.2	8.4
	5% WB	MCDBR	10.9	10.6	12.9	19.5	19.0	19.3	17.3	17.4	18.2	14.1	10.1	9.7
		MCWBR	10.1	9.6	9.8	11.9	10.3	8.7	7.6	8.3	9.8	8.8	7.7	8.6
(40) Clear-Sky Solar Irradiance	taub		0.284	0.317	0.364	0.394	0.407	0.405	0.427	0.408	0.371	0.350	0.314	0.285
	taud		1.996	1.980	1.924	2.069	2.201	2.293	2.236	2.265	2.318	2.214	2.019	1.943
	Ebn at Noon		180	223	243	258	263	266	256	253	246	214	163	140
	Edh at Noon		21	32	44	45	42	39	40	36	30	26	21	17
(44) All-Sky Solar Radiation	RadAvg		242	546	964	1359	1568	1878	1689	1339	864	440	217	142
	RadStd		30	49	80	96	173	164	107	116	111	40	32	20

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page